Supplementary material

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1 Asymptotic convergence of the phase-domain mapping

The phase-domain coordinate mapping simplifies the estimation architecture by evaluating the accumulated phase over a deterministic temporal window Δt_k, relaxing the continuous path-survival condition required by a strict first-passage time (FPT) formulation. We demonstrate here that under the physiological constraints of the sinoatrial node, this terminal-threshold formulation converges asymptotically to the boundary-crossing probability, making the analytical discrepancy negligible

To formalize this, consider the complete, closed-form cumulative distribution function (CDF) for the first-passage time τ of a one-dimensional drift-diffusion process with constant net drift ν, volatility σ, and an absorbing boundary at $a = 1.0$:

$$F_\tau(t) = \Phi\left(\frac{\nu t - 1.0}{\sigma\sqrt{t}}\right) + \exp\left(\frac{2\nu}{\sigma^2}\right)\Phi\left(\frac{-\nu t - 1.0}{\sigma\sqrt{t}}\right)$$

where $\Phi(\cdot)$ denotes the standard normal cumulative distribution function. The total probability density is governed by two mathematically distinct components.

The total probability density is governed by two mathematically distinct components derived via the reflection principle of Brownian motion. First, the primary drift-diffusion term $\Phi\left(\frac{\nu t-1.0}{\sigma\sqrt{t}}\right)$, which represents the unconstrained probability that the continuous process is located above the threshold at the temporal slice t. This term maps directly to the linear Gaussian observation equation utilized in our phase-domain inversion. Second, the boundary reflection term $\exp\left(\frac{2\nu}{\sigma^2}\right)\Phi\left(\frac{-\nu t-1.0}{\sigma\sqrt{t}}\right)$, which serves as a mathematical penalty accounting exclusively for the sub-population of sample paths that crossed the absorbing boundary prior to time t but subsequently diffused back below the threshold.

The mathematical validity of omitting the boundary-reflection correction depends entirely on the biological regime of the physical system. In cardiac electrophysiology, the intrinsic depolarization of pacemaker cells is dictated by a massive, highly directed ionic current relative to microscopic thermodynamic perturbations. Consequently, the sinoatrial node operates within a high-drift, low-noise regime, characterized by a large dimensionless drift-to-diffusion ratio ($\mathrm{Pe} = \frac{\nu}{\sigma} \gg 1$, where σ is the diffusion coefficient at unit threshold).

Under these physiological parameters, we evaluate the behavior of the correction term. Because both the net drift rate ν and the normalized threshold are positive, the argument of the normal CDF in the reflection term is deeply negative for all valid temporal intervals:

$$\lim_{\frac{\nu}{\sigma}\to\infty} \frac{-\nu t - 1.0}{\sigma\sqrt{t}} = -\infty$$

Because the tail of the standard normal distribution decays exponentially ($\Phi(-x) \sim \frac{1}{x\sqrt{2\pi}}e^{-x^2/2}$ as $x \to \infty$), the evaluation of $\Phi\left(\frac{-\nu t-1.0}{\sigma\sqrt{t}}\right)$ approaches zero asymptotically. Even when scaled by the preceding exponential martingale multiplier $\exp\left(\frac{2\nu}{\sigma^2}\right)$, the product is dominated entirely by the super-exponential decay of the normal tail.

For standard physiological intervals ($\Delta t_k \sim 1.0$ s) and empirical autonomic volatilities observed in mammalian cardiac pacing, the absolute numerical value of this second term sits safely below the machine epsilon limit ($\approx 2.22 \times 10^{-16}$) of modern double-precision floating-point representations.

Therefore, the true first-passage distribution collapses cleanly into the single Gaussian component:

$$F_\tau(t) \approx \Phi\left(\frac{\nu t - 1.0}{\sigma\sqrt{t}}\right)$$

By establishing that the probability of a path crossing the depolarization threshold and returning below it within a single cardiac cycle is biologically and numerically negligible, the phase-domain inversion ceases to be a heuristic approximation. It represents the physical limit of the continuous point process, safely vindicating the use of closed-form linear-Gaussian Kalman filtering theory.

2 Estimation algorithm

This section details the specific algorithmic execution and numerical safeguards required to implement the augmented phase-domain state-space model. The computational architecture is divided into the continuous wavelet extraction algorithm, the numerically stable evaluation of the continuous-time transitions, the geometric enforcement of the Kalman covariance updates, and the structural assembly of the unconstrained optimization space. The continuous-discrete filtering architecture evaluates the augmented state vector $\tilde{X}(t) = [x_p(t), x_s(t), I_p(t), I_s(t)]^T$. For each discrete observation interval Δt_k, the recursive estimation proceeds as follows:

2.1 Deterministic wavelet anchoring and data-driven priors

The extraction of the deterministic spectral anchors and kinetic priors relies on the continuous Morlet wavelet transform. To perform this integration over a continuous time-frequency domain, the uneven inter-beat interval sequence is first inverted to an instantaneous heart rate and interpolated onto an isochronous grid sampled at four Hertz using a cubic spline. The interpolated signal is mean-centered to prevent the direct current component from dominating the low-frequency wavelet scales. The biwavelet algorithm evaluates the time-frequency surface, which is subsequently averaged across the temporal dimension to yield the global wavelet spectrum. Because the wavelet scales are distributed logarithmically, a direct summation of the spectral bins induces a frequency bias. The absolute power within the low-frequency and high-frequency physiological bands is evaluated as a Riemann sum, scaled by the discrete frequency differential:

$$P_{band} = \sum_{f \in band} \bar{W}(f) df$$

where $\bar{W}(f)$ is the global wavelet power at frequency f, and df is the absolute difference between adjacent frequency scales. Because the continuous model utilizes an orthogonal parameterization that disentangles the amplitude of the biological drive from its memory, the steady-state variance ratio of the parasympathetic to sympathetic branch, V_p, is fixed directly to the empirical wavelet energy quotient:

$$V_p = \frac{P_{HF}}{P_{LF}}$$

To inform the temporal behavior of the system, the algorithm further extracts the spectral bandwidth of the physiological drives. The frequency dispersion around the centroid of each band is calculated as the power-weighted spectral standard deviation. Because the continuous clearance rate (κ) of a first-order autoregressive process is mathematically equivalent to its angular half-power bandwidth, these empirical dispersions are multiplied by 2π to serve as the patient-specific modes for the Bayesian regularization priors applied during numerical optimization.

2.2 Evaluation of the continuous-time transitions

Evaluating the continuous-time transitions of the Ornstein-Uhlenbeck processes over discrete, uneven observation intervals requires computing the exact matrix exponential and the corresponding process covariance integrals. To guarantee that the variance geometry remains conserved as the numerical optimizer tests different clearance rates, the continuous diffusion coefficient is evaluated dynamically within the integration step as $\sigma^2 = 2\kappa V$, where V is the stationary amplitude anchor.

When the product of the neural clearance rate κ and the interval duration Δt_k approaches zero, the analytical solutions become computationally unstable. Standard evaluation of the integrated phase variance involves operations configured as $(1 - \exp(-x))/x$, which succumb to catastrophic floating-point cancellation limits in standard architectures. To resolve this instability, the algorithm evaluates the dimensionless argument $x = \kappa \Delta t_k$. For values of $x \geq 10^{-3}$, the integrated phase covariance $Q_{3,3}$ is evaluated using the closed-form continuous integral:

$$Q_{3,3} = \frac{2\kappa V}{\kappa^2} \left[\Delta t_k - \frac{2}{\kappa}(1 - e^{-\kappa \Delta t_k}) + \frac{1}{2\kappa}(1 - e^{-2\kappa \Delta t_k}) \right]$$

When the observation interval or the clearance rate causes $x < 10^{-3}$, the algorithm transitions to a truncated Taylor series expansion for the marginal components of the process covariance matrix Q_k. This substitution bypasses the exponential subtractions entirely, ensuring that the process covariance matrices remain positive semi-definite near the limit boundary:

$$Q_{3,3} \approx 2\kappa V \left[\frac{\Delta t_k^3}{3} - \frac{\kappa \Delta t_k^4}{4} + \frac{7\kappa^2 \Delta t_k^5}{60} \right]$$

2.3 Kalman covariance updates

The dual-filter Generalized Least Squares engine relies on two concurrent state vectors to compute the normalized innovations. The primary vector $X^{(0)}$ evaluates the stochastic progression assuming a null intrinsic baseline, while the secondary vector $X^{(\nu)}$ evaluates the deterministic system response to the time step variable. The numerical stability of this recursive filter is maintained through the specific algebraic formulation of the covariance update. Standard Kalman implementations utilize the subtraction form for the a posteriori covariance, which can lose positive semi-definiteness under conditions of low measurement noise or prolonged estimation. The present architecture utilizes the Joseph form of the covariance update equation:

$$P_{k|k} = (I - K_k H) P_{k|k-1} (I - K_k H)^T + K_k R_k K_k^T$$

where K_k is the Kalman gain, H is the observation matrix, and R_k is the total observation variance, defined as $R_k = \lambda_R \Delta t_k + H Q_k H^T$, where Q_k is the propagated process covariance of the integrated autonomic states.

2.3.1 Biophysical partitioning of system variance

Within this structural formulation, it is mathematically necessary to distinguish the specific physical sources of system variance. The continuous process covariance matrix (Q_k) explicitly captures the integrated Brownian perturbations of the ionic currents accumulating throughout the depolarization interval. In contrast, the discrete observation noise variance (R_k) absorbs instantaneous stochasticity at the absorbing boundary. Specifically, R_k quantifies the biophysical jitter of the cellular depolarization threshold itself, as well as the deterministic quantization error inherent to the finite sampling frequency of the electrocardiogram hardware.

This algebraic structure guarantees that the covariance matrix $P_{k|k}$ remains symmetric and positive semi-definite regardless of the numerical magnitude of the measurement updates. Explicit structural symmetry is subsequently enforced by averaging the matrix with its own transpose.

2.3.2 Hybrid continuous-discrete jump system and phase repolarization

Crucially, the SPARK estimation architecture operates as a hybrid continuous-discrete jump system. Following the measurement update at each heartbeat, the biological repolarization of the sinoatrial node is geometrically enforced by instantaneously resetting the integrated phase states (I_p, I_s) and their corresponding dimensions within the posterior covariance matrix $(P_{k|k})$ to zero. Conversely, the latent autonomic states (x_p, x_s) and their covariances are preserved across the boundary. This asymmetric reset mechanism ensures that the continuous, trans-beat memory of the autonomic nervous system is maintained while preventing artificial, non-physiological phase accumulation across sequential intervals.

At the conclusion of the filtering sequence of length N, the optimal baseline intrinsic drift $\hat{\nu}_0$ and the global system volatility $\hat{\sigma}^2_{sys}$ are computed using the stored arrays of the stochastic innovations $v^{(0)}$, the deterministic responses $v^{(\nu)}$, and the sequential innovation variances S:

$$\hat{\nu}_0 = \frac{\sum_{k=1}^{N}\left(v_k^{(0)} v_k^{(\nu)}/S_k\right)}{\sum_{k=1}^{N}\left((v_k^{(\nu)})^2/S_k\right)}$$

$$\hat{\sigma}^2_{sys} = \frac{1}{N}\sum_{k=1}^{N}\frac{(v_k^{(0)} - \hat{\nu}_0 v_k^{(\nu)})^2}{S_k}$$

The final physiological state geometry is reconstructed through the linear superposition of the deterministic and stochastic storage matrices. The extracted baseline scales the deterministic response, yielding the uncoupled biological states:

$$\hat{X}_{final} = X^{(0)} - \hat{\nu}_0 X^{(\nu)}$$

2.4 Unconstrained optimization space

The numerical parameter estimation relies on a bounded Broyden-Fletcher-Goldfarb-Shanno optimization algorithm. The algorithm searches an unconstrained three-dimensional Euclidean space, denoted by the vector θ. The elements of θ are mapped to the physical parameters through exponential transformations. The sympathetic clearance rate is mapped as $\kappa_s = \exp(\theta_1)$. The structural identifiability constraint dictating that vagal kinetics must operate faster than sympathetic kinetics is enforced via the mapping $\kappa_p = \kappa_s + \exp(\theta_2)$. The fractional threshold noise is mapped as $\lambda_R = \exp(\theta_3)$.

The objective function evaluates the negative regularized log-posterior. The profiled log-likelihood $\mathcal{L}^*$ is formulated exclusively from the innovation variance and the analytically derived global system volatility:

$$\mathcal{L}^* = -\frac{1}{2}\sum_{k=1}^{N}\ln(S_k) - \frac{N}{2}\ln(\hat{\sigma}^2_{sys})$$

Conjugate probability bounds are applied as additive algebraic penalties to the profiled log-likelihood. The continuous polynomial curves for the clearance rates are structured as Gamma priors, explicitly centered on the data-driven bandwidth modes extracted from the wavelet spectrum, tying the regularized search space directly to the patient's individual physiology. An Inverse-Gamma prior is similarly applied to penalize excessive allocations of variance to the threshold noise fraction. The final cost function passed to the numerical optimizer is the inversion of this regularized topology:

$$\text{Cost} = -\mathcal{L}^* - [(\alpha_s - 1)\ln(\kappa_s) - \beta_s\kappa_s] - \left[(\alpha_p - 1)\ln(\kappa_p) - \beta_p\kappa_p\right] - \left[-(a+1)\ln(\lambda_R) - \frac{b}{\lambda_R}\right]$$

3 Goodness-of-fit diagnostic in diffusion-based phase-domain architectures

The validation of point-process models using the Kolmogorov-Smirnov (KS) goodness-of-fit test relies heavily on the time-rescaling theorem. However, it is mathematically necessary to distinguish between the stochastic assumptions of classical time-rescaling theorem (which assumes probabilistic event arrivals) and the SPARK framework (which models deterministic thresholds over stochastic diffusion paths). This section formally demonstrates that while the underlying generative noise distributions differ, the probability integral transform utilized in both frameworks ensures isomorphic equivalence for goodness-of-fit evaluation.

3.1 The classical poisson arrival formulation

In classical point-process statistics, an event (heartbeat) is modeled as a stochastic arrival governed by a conditional intensity function $\lambda(t)$. Under this assumption, the generation of events follows an inhomogeneous Poisson process.

The classical time-rescaling theorem states that if the continuous intensity function $\lambda(t)$ is integrated over the observed inter-beat interval $[t_{k-1}, t_k]$, the resulting integrated intensity Λ_k transforms the uneven temporal sequence into a homogeneous Poisson process with a unit rate:

$$\Lambda_k = \int_{t_{k-1}}^{t_k} \lambda(t)dt$$

Because the underlying generative noise of a unit-rate Poisson process is exponentially distributed, $\Lambda_k \sim \text{Exp}(1)$. The theorem executes a probability integral transform using the exponential cumulative distribution function (CDF) to map the sequence into a uniform distribution:

$$u_k = 1 - \exp(-\Lambda_k) \sim \mathcal{U}(0,1)$$

This $\mathcal{U}(0,1)$ sequence is then evaluated geometrically using the KS test.

3.1.1 The phase-domain diffusion formulation

In contrast, the SPARK framework abandons the Poisson arrival assumption. Cardiac pacing is modeled explicitly as a continuous-time Integrate-and-Fire (IF) diffusion process. Here, the event is triggered deterministically when the cellular membrane potential reaches a threshold of 1. The stochasticity arises exclusively from the continuous Brownian motion (Wiener process, dW_t) of the biological environment during the interval.

The governing stochastic differential equation over the observed interval is defined as:

$$\int_{t_{k-1}}^{t_k} \lambda(t)dt + \int_{t_{k-1}}^{t_k} \sigma dW_t = 1$$

In the inverted phase-domain filter, the deterministic drift component represents the predicted phase accumulation:

$$\hat{\phi}_{k|k-1} = \int_{t_{k-1}}^{t_k} \lambda(t)dt$$

By observing the threshold ($y_k = 1$), the filter innovation (v_k) isolates the stochastic noise component:

$$v_k = 1 - \hat{\phi}_{k|k-1} = \int_{t_{k-1}}^{t_k} \sigma dW_t$$

Crucially, because the integral of a Wiener process yields a Gaussian distribution by definition, the isolated innovation v_k does not follow an exponential distribution. Instead, it is strictly normal, characterized by a mean of zero and a calculable innovation variance S_k:

$$v_k \sim \mathcal{N}(0, S_k)$$

3.1.2 Isomorphic equivalence of the time-rescaling theorem principle

To execute the goodness-of-fit evaluation under the diffusion paradigm, the innovation is standardized to extract the pure standard normal generative noise:

$$\epsilon_k = \frac{v_k}{\sqrt{S_k}} \sim \mathcal{N}(0,1)$$

We then apply the same mathematical mechanism utilized in the classical time-rescaling theorem: the probability integral transform. However, because the generative noise of the SPARK diffusion model is standard normal rather than exponential, the transformation requires the standard normal CDF, $\Phi(\cdot)$:

$$u_k = \Phi(\epsilon_k) \sim \mathcal{U}(0,1)$$

Although the SPARK architecture operates under diffusion rather than Poisson assumptions, the fundamental mechanism of the time-rescaling theorem, isolating the true generative noise and mapping it to a uniform distribution via its probability integral transform, remains mathematically exact. Consequently, testing the sequence $u_k = \Phi(\epsilon_k)$ for uniformity via the Kolmogorov-Smirnov distance constitutes the time-rescaling principle for first-passage time point-process models.

4 Emergent macroscopic heteroskedasticity via first-passage time

A well-documented phenomenon in cardiorespiratory physiology is vagal gating (respiratory sinus arrhythmia), wherein high parasympathetic tone dynamically amplifies the macroscopic variance of the observed interval sequence. Standard state-space architectures struggle to capture this without introducing mathematically intractable, state-dependent observation noise matrices ($R_k \propto X_{PNS}$).

The SPARK framework resolves this natively through its isomorphic coordinate mapping. Within the continuous phase domain, the system is modeled with constant, homoskedastic diffusion (σdW_t), ensuring tractability within the linear-Gaussian generalized least squares architecture. However, the observable discrete heartbeat τ_k is the first-passage time of this process reaching the absorbing boundary $a = 1.0$.

For a Brownian motion with local net drift ν_k and constant volatility σ, the first-passage time follows an Inverse Gaussian distribution. While the expected macroscopic interval is $\mathbb{E}[\tau_k] = 1.0/\nu_k$, the analytical variance of the observed interval is non-linear with respect to the drift:

$$\mathrm{Var}(\tau_k) = \frac{\sigma^2}{\nu_k^3}$$

Because the macroscopic variance is inversely proportional to the cube of the drift (ν_k^3), the framework guarantees heteroskedasticity in the time domain. When parasympathetic tone increases, the instantaneous net drift ν_k is suppressed. Even with constant underlying system noise (σ^2), this suppression induces an exponential expansion in $\mathrm{Var}(\tau_k)$, perfectly mimicking biological vagal gating. Conversely, parasympathetic withdrawal increases ν_k, driving the macroscopic variance toward zero and recovering the rigid, low-variance pacing characteristic of pure sympathetic drive.

5 Continuous-discrete state-space formulation via matrix exponential

A core operational advantage of the SPARK framework is its ability to evaluate continuous-time kinetics across variable discrete sampling intervals without relying on Euler-Maruyama approximations. This is achieved by defining the autonomic state vector such that it captures both the instantaneous neural drives and their accumulated path integrals between heartbeats.

Let the continuous-time autonomic state vector be $\mathbf{X}(t) = [x_p(t), x_s(t), I_p(t), I_s(t)]^\top$, where $x_p(t)$ and $x_s(t)$ represent the instantaneous parasympathetic and sympathetic Ornstein-Uhlenbeck processes, and $I_p(t)$ and $I_s(t)$ represent their running integrals.

The continuous-time system dynamics are governed by the linear SDE:

$$d\mathbf{X}(t) = \mathbf{A}\mathbf{X}(t)dt + \mathbf{G}d\mathbf{W}(t)$$

where $\mathbf{A}$ is the continuous system evolution matrix defined by the kinetic clearance rates (κ_p, κ_s):

$$\mathbf{A} = \begin{bmatrix} -\kappa_p & 0 & 0 & 0 \\ 0 & -\kappa_s & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

Because this system is linear time-invariant between observations, the discrete-time transition over any arbitrary stochastic inter-beat interval Δt_k is solved analytically via the matrix exponential:

$$\mathbf{F}_k = \exp(\mathbf{A}\Delta t_k) = \begin{bmatrix} e^{-\kappa_p \Delta t_k} & 0 & 0 & 0 \\ 0 & e^{-\kappa_s \Delta t_k} & 0 & 0 \\ \frac{1-e^{-\kappa_p \Delta t_k}}{\kappa_p} & 0 & 1 & 0 \\ 0 & \frac{1-e^{-\kappa_s \Delta t_k}}{\kappa_s} & 0 & 1 \end{bmatrix}$$

This transition matrix $\mathbf{F}_k$ allows the Kalman filter to propagate the state vector continuously across non-uniform biological time. Upon the registration of a macroscopic interval (the phase-domain boundary crossing), the integral states I_p and I_s are projected into the observation equation, and subsequently reset to zero. This mathematical reset perfectly mirrors the biophysical depolarization and refractory resetting of the sinoatrial node, preserving absolute continuous-time fidelity without accumulation of discrete-time discretization errors.

5.1 Explicit derivation of the transition matrix

To demonstrate the structural origin of this transition matrix, we evaluate $\exp(\mathbf{A}\Delta t_k)$ using the fundamental Taylor series definition of the matrix exponential:

$$\exp(\mathbf{A}\Delta t_k) = \sum_{n=0}^{\infty} \frac{(\mathbf{A}\Delta t_k)^n}{n!} = \mathbf{I} + \mathbf{A}\Delta t_k + \frac{1}{2!}\mathbf{A}^2 \Delta t_k^2 + ...$$

Given the specific block structure of the continuous system matrix $\mathbf{A}$, evaluating its higher-order matrix powers reveals a strict deterministic pattern. For any integer $n \geq 1$:

$$\mathbf{A}^n = \begin{bmatrix} (-\kappa_p)^n & 0 & 0 & 0 \\ 0 & (-\kappa_s)^n & 0 & 0 \\ (-\kappa_p)^{n-1} & 0 & 0 & 0 \\ 0 & (-\kappa_s)^{n-1} & 0 & 0 \end{bmatrix}$$

Substituting this matrix pattern back into the infinite series expansion allows us to evaluate the discrete elements of the transition matrix independently. For the top-left diagonal element governing the instantaneous parasympathetic state (x_p), the series collapses directly into the standard scalar exponential function:

$$\sum_{n=0}^{\infty} \frac{(-\kappa_p \Delta t_k)^n}{n!} = e^{-\kappa_p \Delta t_k}$$

For the bottom-left off-diagonal element governing the cumulative integral state (I_p), the series expansion operates exclusively for $n \geq 1$ (as the $n = 0$ identity matrix $\mathbf{I}$ has a zero in this position). By extracting a factor of $-1/\kappa_p$ from the terms, we reconstruct the exponential series:

$$\sum_{n=1}^{\infty} \frac{\Delta t_k^n}{n!} (-\kappa_p)^{n-1} = -\frac{1}{\kappa_p} \sum_{n=1}^{\infty} \frac{(-\kappa_p \Delta t_k)^n}{n!}$$

Recognizing that the right-side summation is simply the complete Taylor series for $e^{-\kappa_p \Delta t_k}$ minus the $n = 0$ term (which equals 1), the expression resolves analytically to the running integral:

$$-\frac{1}{\kappa_p} \left(e^{-\kappa_p \Delta t_k} - 1 \right) = \frac{1 - e^{-\kappa_p \Delta t_k}}{\kappa_p}$$

Because the parasympathetic and sympathetic pathways are mathematically decoupled within the continuous system matrix (i.e., they do not cross-talk), the same algebraic reduction applies symmetrically to the sympathetic parameters. The instantaneous sympathetic state at position (2,2) evaluates independently to:

$$\sum_{n=0}^{\infty} \frac{(-\kappa_s \Delta t_k)^n}{n!} = e^{-\kappa_s \Delta t_k}$$

Correspondingly, its cumulative integral state at position (4,2) resolves to:

$$-\frac{1}{\kappa_s} \left(e^{-\kappa_s \Delta t_k} - 1 \right) = \frac{1 - e^{-\kappa_s \Delta t_k}}{\kappa_s}$$

Finally, because the entire right half of the block matrix $\mathbf{A}^n$ is identically zero for all $n \geq 1$, the integral preservation states at positions (3,3) and (4,4) are populated solely by the $n = 0$ identity matrix $\mathbf{I}$, thereby remaining 1.

Assembling these independent scalar solutions back into the full 4×4 geometry yields the final closed-form transition matrix $\mathbf{F}_k$. This derivation explicitly proves that the accumulation of both the parasympathetic and sympathetic autonomic integral states is evaluated natively across continuous time, entirely bypassing the discretization errors inherent to numerical integration schemes (e.g., Euler or Runge-Kutta).